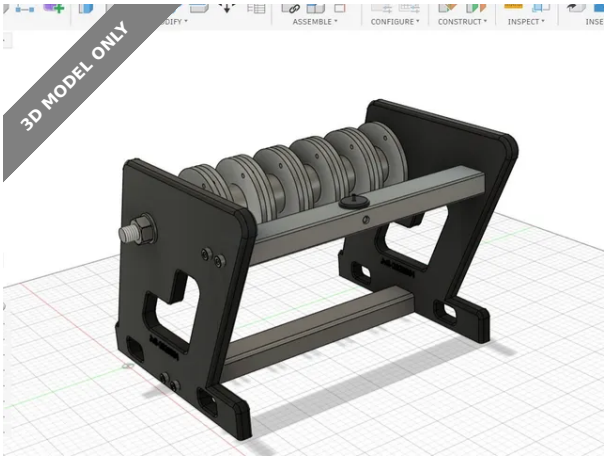


Multi-strand wire dispenser for parallel motor/ stator/rotor winding

koBot

VIEW IN BROWSER

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Summary

A simple and reliable multi-strand wire dispenser that makes parallel stator winding faster and cleaner.

Hobby & Makers > RC & Robotics

Tags: stator winding actuator bldc dispenser diy enamel magnet motor parallel robotics tool wire workshop qdd minicheetah

This is a compact 3D-printable multi-strand wire dispenser designed to simplify parallel coil winding for BLDC stators and similar motor projects.

It keeps multiple enamel wires aligned and tensioned, making it easier to wind clean and consistent coils without tangling. The design is especially useful for 6-in-hand winding as commonly used in high-torque robotic actuators such as Mini-Cheetah-style QDD motors.

The dispenser is modular, easy to print, and requires supports only for the spool retaining rings.

Tested with 6 × 26 AWG enamel copper wire.

Ideal for:

- BLDC motor winding
- stator rewinding
- robotics actuator builds
- transformer experiments
- parallel enamel wire handling

Designed for reliability, repeatability, and clean workshop workflow.

Print settings

- Recommended material: PLA or PETG
- Layer height: 0.2 mm
- Infill: 15–25%
- Supports needed for printing the six spools

Model files



crossbar.stl



spool.stl



lid.stl



side_plate.stl



wire_guide.stl

Other files



readme.txt

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